RITWIK SINGH

Computational Biology | Epigenomics, Network Analysis & Biomedical Imaging | New Delhi
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RESEARCH SUMMARY

Computational biologist (M.Sc. Biological Sciences + B.E. Electronics and Instrumentation, BITS Pilani) working on transcriptomic and epigenomic network analysis in clinical *Plasmodium falciparum* isolates. My longer-term aim is to combine diagnostic scanning imagery with AI-driven analysis thereby linking quantitative imaging signatures to molecular phenotypes to guide targeted therapeutic strategies.

EDUCATION

BITS Pilani - Pilani, Rajasthan — July 2025

M.Sc. Biological Sciences* | *B.E. Electronics and Instrumentation *(Integrated Dual Degree)*

- **CGPA**: 8.74 / 10
- ***Master's Thesis****: "Correlation of Gene Expression with Epigenetic Modifications in Plasmodium falciparum" - supervised by* ***Senior Professor Ashis Kumar Das****, Department of Biological Sciences*
- ***Design Project****: Feasibility Analysis of Rotatable Solar Panels - demonstrated a ~20% average rise in energy production using real-time field data from Rajasthan - overseen by* ***Professor Hitesh Dutt Mathur****, Department of Electrical and Electronics Engineering*
- ***MCN Scholarship****: renewable semester-wise grant for demonstrated academic excellence*

Sunbeam English School - Varanasi, Uttar Pradesh — May 2020

CBSE Senior Secondary Examination

- *97.2% (Mathematics 100/100)*
- *Ranked in the top 0.1% of test takers nationally*

NOTABLE ACADEMIC WORK

Correlation of Gene Expression with Epigenetic Modifications in *P. falciparum*

On-campus Thesis — ***Jan 2025 – May 2025***

- *Characterised DNA methylation profiles across clinical replicates and summarised methylation patterns by sample.*
- *Identified and annotated DMRs stratified by disease manifestation.*
- *Quantified the correlation between methylation state and gene expression at a genome level.*
- *Assembled a hub-gene list from the integrated analysis and proposed candidate intervention points for next-generation antimalarial strategies.*

Transcriptomic and Epigenetic Network Analysis in *Plasmodium falciparum*

Special Project & Lab-Oriented Projects — ***May 2023 – May 2024***

- *Constructed weighted gene co-expression networks (WGCNA) from microarray expression data of clinical P. falciparum isolates, generating independent sense and antisense networks - an approach that surfaces regulatory structure missed by conventional sense-only analysis.*
- *Identified candidate hub genes from network topology across clinical samples spanning multiple disease manifestations.*
- *Extended the framework to DNA methylation, mapping differentially methylated regions by manifestation and correlating methylation against expression to propose intervention points for next-generation antimalarials.*

PUBLICATIONS

Singh, R. "Gene co-expression network analysis of sense and antisense transcriptomes from *Plasmodium falciparum* clinical isolates" Manuscript in preparation; expected completion December 2026.

ENGINEERING & INDUSTRY EXPERIENCE

Uber - Software Engineering Intern

Hyderabad, Telangana ***July 2024 – Dec 2024***

- *Worked within the Payments Compliance team on global regulatory requirements while minimising user-facing friction.*
- *Optimised the Level-1 KYC onboarding flow for Uber Money users across the UK and EU.*
- *Designed and ran A/B tests prior to full-scale launch and quantified the impact on onboarding completion.*
- *Built the flow for straightforward extension to additional geographies including Mexico, New Zealand, and Japan.*

Reliance Jio Platforms - Web Development Intern

Mumbai, Maharashtra ***May 2022 – July 2022***

Built a ReactJS front-end for a data visualisation service rendering user-uploaded CSV data across stacked area, percent area, bar, pie, radar, box, candlestick, and bean plots.

TEACHING

Teaching Assistant

Microprocessors and Interfacing ***Jan 2024 – May 2024***

AWARDS | SCHOLARSHIPS | CERTIFICATIONS

Apoorv Foundation Scholarship and V. Ramakrishna Award

awarded to the school's top-ranked science student and to the school's best Class 12 CBSE performer respectively

SAT India Top Performer, 2020

Rank 158, SOF International Mathematics Olympiad

Winner, Annual COFAS International, City Montessori School (2016)

Introduction to Intellectual Property - ***University of Pennsylvania (Coursera), Oct 2025***

LEADERSHIP AND OUTREACH

IT Consultant

Pragati Public School, Varanasi ***Nov 2020 – April 2023***

Built a Moodle-based LMS to sustain online teaching through the COVID-19 pandemic and its aftermath; redesigned the school website and managed its social media presence; prepared and filed the school's ATL tranche application, securing fund release from NITI Aayog under the Atal Innovation Mission.

BOSM Coordinator

Hindi Press Club, BITS Pilani ***Aug 2023 – Dec 2023***

Organised open-mic commentary for basketball and cricket fixtures; oversaw drafting and publication of 200 Hindi newsletters daily across the four-day annual sports festival.

INTERESTS

AI applications in non-conventional domains | Writing on technology in public systems| Schooling and pedagogy | Natural and historical documentaries | Trekking and wildlife exploration | Cricket as an off-spin bowler and middle-order batsman

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TECHNICAL SKILLS

Sequencing & QC	FastQC, MultiQC, read trimming, BEDtools
Transcriptomics	Microarray preprocessing and normalisation differential expression, sense–antisense transcript quantification, Intra-Array Clustering (IAC)
Network Biology	WGCNA, module–trait correlation, hub-gene identification, network topology analysis, Cytoscape Functional Analysis, PlasmoDB, NCBI/GEO data retrieval
Programming	R (tidyverse, Bioconductor), Python (pandas, NumPy, scikit-learn), Bash
Engineering & Web	Signal processing and instrumentation, ReactJS, Git, GitHub, A/B testing and experiment design, Go